Stoichiometry and limiting reactants

Balance first, then compare available reaction batches and calculate product.

N2 + 3 H2 -> 2 NH3

Particle example: 4 N2 + 9 H2



Reaction batches available	Conclusion
N2: 4/1 = 4; H2: 9/3 = 3	H2 limits. Three full batches form 6 NH3; 1 N2 remains.
General rule	Divide each available mole amount by its coefficient; the smallest possible batch count limits product.

Mass example: 14.0 g N2 + 6.00 g H2

N2 permits: (14.0 g / 28.02 g/mol) x 2 = 0.9993 mol NH3.
H2 permits: (6.00 g / 2.016 g/mol) x (2/3) = 1.984 mol NH3.
N2 limits: 0.9993 mol x 17.03 g/mol = 17.0 g NH3.

Percent yield = actual / theoretical x 100%

If 12.8 g is collected: 12.8/17.0186 x 100 = 75.2%. Keep extra digits until the final answer. Balance coefficients without changing subscripts; compare mole ratios, not masses alone. Complete-reaction teaching model; real ammonia equilibrium is omitted.

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Self-check questions and explanations. Refer to the visual guide on the front.

What is the balanced equation for ammonia formation?

N2 + 3 H2 -> 2 NH3. One N2 contributes 2 N atoms; three H2 contribute 6 H atoms, matching two NH3 molecules.

With 4 N2 and 9 H2 molecules, what forms and what remains?

Three complete reaction batches consume 3 N2 and 9 H2, producing 6 NH3. One N2 molecule remains and no H2 remains. Atoms must be conserved in the drawing.

Why is H2 limiting despite its larger molecule count?

H2 limits. Need 3 H2 per N2, so available batch counts are 4/1 = 4 versus 9/3 = 3; the smaller batch count controls product.

With 2.00 mol N2 and 3.00 mol H2, how much NH3 forms?

H2 limits: 3.00 mol H2 x (2 mol NH3 / 3 mol H2) = 2.00 mol NH3. N2 used is 1.00 mol; N2 remaining is 1.00 mol.

With 14.0 g N2 and 6.00 g H2, what is the theoretical NH3 mass?

N2 gives 14.0/28.02 x 2 = 0.9993 mol NH3; H2 could give 6.00/2.016 x 2/3 = 1.984 mol. N2 limits. Theoretical mass = 0.9993 x 17.03 = 17.0 g NH3 (3 significant figures).

If that mass example produces 12.8 g NH3, what is percent yield?

12.8 g / 17.0186 g x 100 = 75.2%. Compare actual recovered product to theoretical maximum. Loss during collection or incomplete conversion could lower yield; identify a mechanism rather than blaming rounding.